

Name:

Matriculation number:

MST-Technologies and Processes

March, 18th, 2019

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points:

60 of 120 in total

Points:

Result:



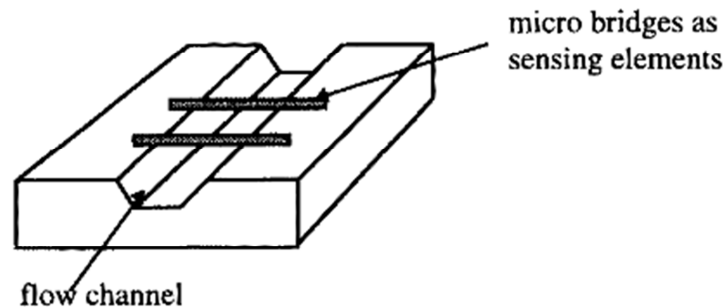
1. Processes**24**

1.1 Processes

6

name 2 processes from each of the following process classes:
transformative, subtractive, additive

- 1.2 Sketch the manufacturing steps required to fabricate the following structure (an anemometric flow sensor)
substrate is silicon; micro bridges are made from platinum



Indicate the wafer type required

2

- 1.3 Sketch the processing for generating the micro bridges (i.e., the electrodes)

8

- 1.4 Sketch the processing for generating the flow channel

8

2. Vacuum technology**25**

- 2.1 What is the Knudsen number and how is it defined?

2

- 2.2 Which are the three relevant Knudsen regimes in vacuum technology?
State the relevant Knudsen number value and the characterize the flow in these regimes

6

- 2.3 Vacuum pumps are usually classified in two categories: name them

2

- 2.4 Classify the following two pumps into the correct category

2

- turbomolecular pump
- cryogenic pump

- 2.5 Sketch and explain the turbomolecular pump

4

- 2.6 Why is a turbomolecular pump not used as a stand-alone system?
How is it generally used?

2

- 2.7 Sketch and explain the cryogenic pump

4

- 2.8 Which second pump system can it be “upgraded” to?

2

- 2.9 Sketch and explain one setup for pressure sensing in vacuum systems

2

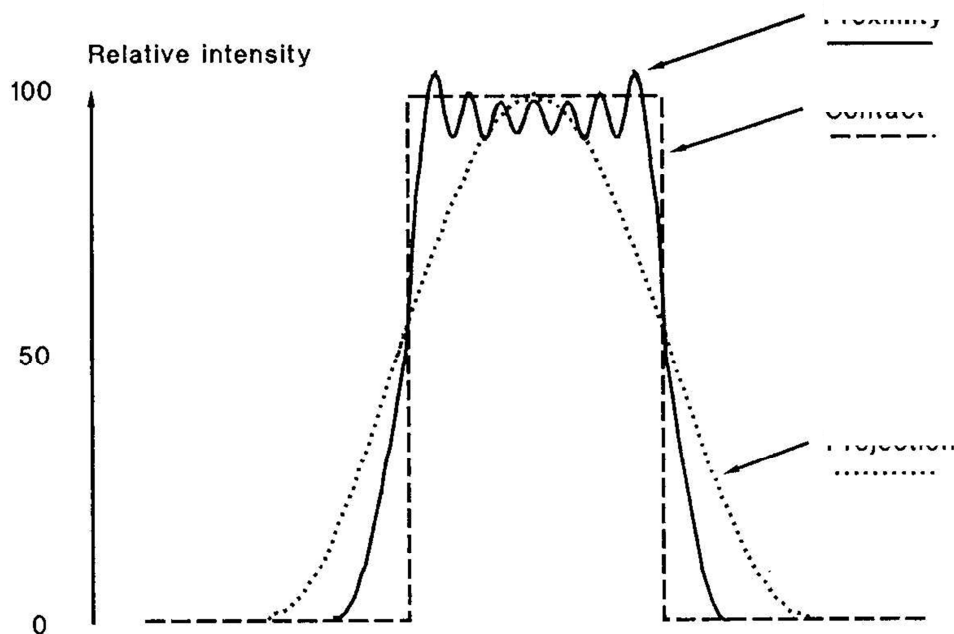


3. Thermal oxidation of silicon**14**

- 3.1 What is the most common model for the thermal oxidation of silicon 2
- 3.2 This model splits the oxidation into two phases: 2
- name them 2
 - state the relation: layer thickness (time) $\rightarrow h(t)$ 2
 - state which physical parameter limits the growth rate 2
- 3.3 There are two major technological pathways and system setups used for thermal oxidation
- name and sketch them 4
 - given typical growth rates 2

4. Photolithography**25**

- 4.1 Sketch and explain the basic lithographic workflow 8
- 4.2 Which two properties of light than can be modulated by the photomask? 2
- 4.3 There are three distinct types of photolithographic setups: sketch and explain them 9
- 4.4 Attribute the following intensity profiles (generated from a single-slit mask) to each of these three setups

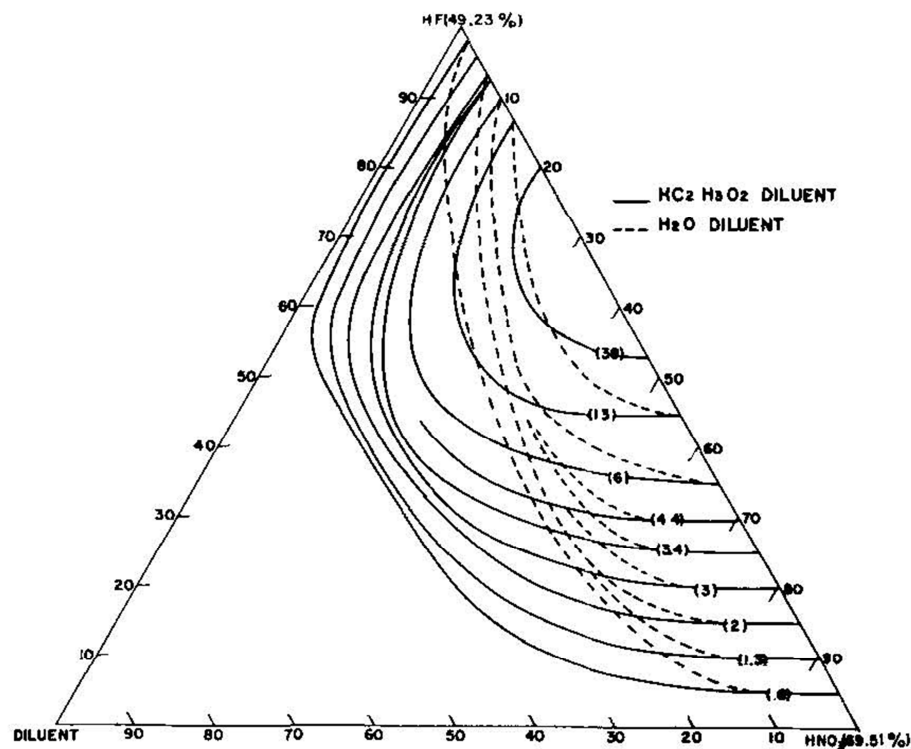


- 4.5 Name the two relevant classes of photoresists 2
- 4.6 Sketch the workflow for both photoresists as well as the final structures when exposing the respective photoresist with a single-slit mask 4



5. Etching**32**

- 5.1 Sketch and explain the following etching technologies 2
- 5.2 • wet etching (sketch one of the systems used for wet etching) 6
- 5.3 • plasma etching 6
- 5.4 • sputter etching 6
- 5.5 State the wet etching reaction of silicon in a mixture of nitric acid and silicon 4
- 5.6 Given the following etching diagram state three etching solutions which give an etching rate of 6 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent 6



- 5.7 Which one would you prefer and why? 2

